

MA1521 Calculus for Computing

Lecture Group 2

Lecturer: Prof To Wing Keung

Lesson 09: Tuesday 08 September 2026

(Week 5)

Remark. Q08.6 to Q08.8 are leftover exercises from the Friday class in Week 4.

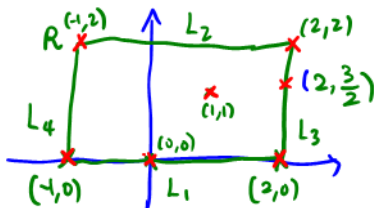
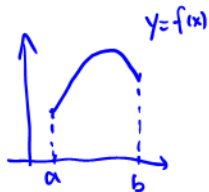
Remark. Q09.1 and Q09.10 are on Sections 13.4, 13.5 of Thomas' Calculus.

Remark. More exercises on Sections 13.4, 13.5, 13.6 and 13.9 will be set for Friday.



Q08.6. Find the maximum and minimum points of the function $f(x, y) = x^2 - 2xy + 2y^2 - 2y$ over the rectangular region $R : -1 \leq x \leq 2, 0 \leq y \leq 2$ in the xy -plane. Hence find the maximum and minimum values of f over the region R .

[Remark: Here you are asked to find the absolute/global maxima and minima of the function f over the region R .



Solution: Given $f(x, y) = x^2 - 2xy + 2y^2 - 2y$ over the region $R : -1 \leq x \leq 2, 0 \leq y \leq 2$.

The closed bounded region R consists of the interior region

$$R^\circ : -1 < x < 2, 0 < y < 2$$

and the boundary, which consists of 4 line segments:

$$L_1 : -1 \leq x \leq 2, y = 0;$$

$$L_2 : -1 \leq x \leq 2, y = 2;$$

$$L_3 : x = 2, 0 \leq y \leq 2;$$

$$L_4 : x = -1, 0 \leq y \leq 2;$$

Note that the endpoints of L_1, L_2, L_3, L_4 are all among the vertices of R : $(-1, 0), (2, 0), (-1, 2), (2, 2)$. (*)

$$f(x, y) = x^2 - 2xy + 2y^2 - 2y.$$

Step 1: Find the critical points of f in the interior R° .

$$f_x = 2x - 2y, \quad f_y = -2x + 4y - 2.$$

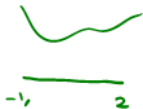
$$\begin{cases} f_x(x, y) = 0 \\ f_y(x, y) = 0 \end{cases} \iff \begin{cases} 2x - 2y = 0, & (1) \\ -2x + 4y - 2 = 0. & (2) \end{cases}$$

From (1), we have $y = x$. Substituting this into (2), we get

$$-2x + 4x - 2 = 0 \implies x = 1.$$

Then we have $y = x = 1$. Thus f has one critical point at $(1, 1)$, which is in R° (since $-1 < 1 < 2$, $0 < 1 < 2$).

$$f(x, y) = x^2 - 2xy + 2y^2 - 2y.$$



Step 2: Find potential extreme points of f along boundary.

(1) Along $L_1 : -1 \leq x \leq 2, y = 0$:

The function reduces to: $g_1(x) = f(x, 0) = x^2$.

For critical points of g_1 , we solve $g_1'(x) = 2x$

$$g_1'(x) = 0 \implies 2x = 0 \implies \underline{x = 0}.$$

Thus potential extreme points of f along L_1 are: $\underline{(0, 0)}$ and the endpoints of L_1 , which are among the vertices in (*).

$$\underline{(-1, 0), (2, 0)}$$

$$f(x, y) = x^2 - 2xy + 2y^2 - 2y.$$

(2) Along $L_2 : -1 \leq x \leq 2, y = 2$:

The function reduces to:

$$g_2(x) = f(x, 2) = x^2 - 4x + 8 - 4 = \underline{x^2 - 4x + 4}.$$

For critical points of g_2 , we solve

$$g_2'(x) = 2x - 4$$

$$g_2'(x) = 0 \implies 2x - 4 = 0 \implies x = 2.$$

Thus potential extreme points of f along L_2 are: (2, 2) and the endpoints of L_2 , which are among the vertices in (*). Note that (2, 2) is also among the vertices in (*).

$$f(x, y) = x^2 - 2xy + 2y^2 - 2y.$$

(3) Along $L_3 : \underline{x = 2}, \underline{0 \leq y \leq 2}$:

The function reduces to:

$$g_3(y) = f(2, y) = 4 - 4y + 2y^2 - 2y = \underline{2y^2 - 6y + 4}.$$

For critical points of g_3 , we solve

$$g'_3(y) = 4y - 6$$

$$g'_3(y) = 0 \implies 4y - 6 = 0 \implies y = \frac{3}{2}.$$

Thus potential extreme points of f along L_3 are: $(2, \underline{\frac{3}{2}})$
and the endpoints of L_3 , which are among the vertices in
(*).

$$f(x, y) = x^2 - 2xy + 2y^2 - 2y.$$

(4) Along $L_4 : x = -1, 0 \leq y \leq 2$:

The function reduces to:

$$g_4(y) = f(-1, y) = 1 + 2y + 2y^2 - 2y = 2y^2 + 1.$$

For critical points of g_4 , we solve

$$g'_4(y) = 4y$$

$$g'_4(y) = 0 \implies 4y = 0 \implies y = 0.$$

Thus potential extreme points of f along L_4 are: $(-1, 0)$ and the endpoints of L_3 , which are among the vertices in (*). Note that $(-1, 0)$ is also among the vertices in (*).

$$f(x, y) = x^2 - 2xy + 2y^2 - 2y.$$

Step 3: Compare the values of f among the potential extreme points from Step 1 and Step 2.

From Steps 1 & 2, the potential extreme points of f are:

$$(1, 1), (0, 0), (2, \frac{3}{2}), (-1, 0), (2, 0), (-1, 2), (2, 2).$$

By direct calculation, we have

$$\begin{aligned} \underline{f(1, 1) = -1}, \quad f(0, 0) = 0, \quad f(2, \frac{3}{2}) = -\frac{1}{2}, \\ f(-1, 0) = 1, \quad f(2, 0) = 4, \quad \underline{f(-1, 2) = 9}, \quad f(2, 2) = 0. \end{aligned}$$

Thus over the region R , f has absolute maximum value **9**, which is attained at the absolute maximum point **$(-1, 2)$** .

Also, over R , f has absolute minimum value **-1** , which is attained at the absolute minimum point **$(1, 1)$** .

Q08.7. Using the method of least squares, find the line (of the form $y = ax + b$) which best fits the three data points $(-1, 1)$, $(1, 2)$ and $(3, 2)$.

Solution: Instead of working out the special case of 3 data points directly, we first set up the general case of n data points $(x_1, y_1), \dots, (x_n, y_n)$. The problem is to minimize the function

$$f(a, b) = \sum_{i=1}^n (ax_i + b - y_i)^2.$$

Then

$$f_a = \sum_{i=1}^n 2(ax_i + b - y_i)x_i, \quad f_b = \sum_{i=1}^n 2(ax_i + b - y_i) \cdot 1.$$

To find critical points of f , we solve

$$\begin{cases} f_a(a, b) = 0 \\ f_b(a, b) = 0 \end{cases} \iff \begin{cases} \sum_{i=1}^n 2(ax_i + b - y_i)x_i = 0, \\ \sum_{i=1}^n 2(ax_i + b - y_i) = 0. \end{cases}$$
$$\iff \begin{cases} a \sum_{i=1}^n x_i^2 + b \sum_{i=1}^n x_i = \sum_{i=1}^n x_i y_i, & (1) \\ a \sum_{i=1}^n x_i + nb = \sum_{i=1}^n y_i. & (2) \end{cases}$$

In this question, we have $n = 3$ with given data points $(-1, 1)$, $(1, 2)$ and $(3, 2)$. Then

$$\sum_{i=1}^n x_i^2 = 11, \quad \sum_{i=1}^n x_i = 3, \quad \sum_{i=1}^n x_i y_i = 7, \quad \sum_{i=1}^n y_i = 5.$$

Then (1) and (2) become

$$\begin{cases} 11a + 3b = 7, & (3) \\ 3a + 3b = 5. & (4) \end{cases}$$

Now (3)-(4) gives

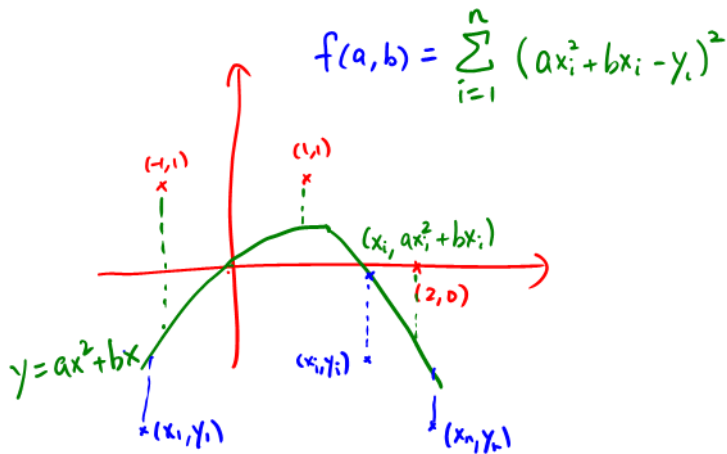
$$8a = 2 \implies a = \frac{1}{4}.$$

Together with (4), we get

$$3b = 5 - 3 \cdot \frac{1}{4} \implies b = \frac{17}{12}.$$

Thus the best fitting line is $y = \frac{1}{4}x + \frac{17}{12}$.

Q08.8. Using the method of least squares, find the quadratic polynomial of the form $y = ax^2 + bx$ which best fits the three data points $(-1, 1)$, $(1, 1)$ and $(2, 0)$.



Solution: Instead of working out the special case of 3 data points directly, we first set up the general case of n data points $(x_1, y_1), \dots, (x_n, y_n)$. The problem is to minimize the function

$$f(a, b) = \sum_{i=1}^n (ax_i^2 + bx_i - y_i)^2.$$

Then

$$f_a = \sum_{i=1}^n 2(ax_i^2 + bx_i - y_i)x_i^2, \quad f_b = \sum_{i=1}^n 2(ax_i^2 + bx_i - y_i) \cdot x_i.$$

To find critical points of f , we solve

$$\begin{cases} f_a(a, b) = 0 \\ f_b(a, b) = 0 \end{cases} \iff \begin{cases} \sum_{i=1}^n 2(ax_i^2 + bx_i - y_i)x_i^2 = 0, \\ \sum_{i=1}^n 2(ax_i^2 + bx_i - y_i) \cdot x_i = 0. \end{cases}$$

$$\iff \begin{cases} a \sum_{i=1}^n x_i^4 + b \sum_{i=1}^n x_i^3 = \sum_{i=1}^n x_i^2 y_i, & (1) \\ a \sum_{i=1}^n x_i^3 + b \sum_{i=1}^n x_i^2 = \sum_{i=1}^n x_i y_i. & (2) \end{cases}$$

In this question, we have $n = 3$ with given data points $(-1, 1)$, $(1, 1)$ and $(2, 0)$. Then

$$\begin{aligned}\sum_{i=1}^n x_i^4 &= 18, & \sum_{i=1}^n x_i^3 &= 8, & \sum_{i=1}^n x_i^2 y_i &= 2, \\ \sum_{i=1}^n x_i^2 &= 6, & \sum_{i=1}^n x_i y_i &= 0.\end{aligned}$$

Then (1) and (2) become

$$\begin{cases} 18a + 8b = 2, & (3) \\ 8a + 6b = 0. & (4) \end{cases}$$

From (4), we have $b = -\frac{4a}{3}$. Together with (3), we get

$$18a - \frac{32a}{3} = 2 \implies a = \frac{3}{11}.$$

Then

$$b = -\frac{4}{3} \cdot \frac{3}{11} = -\frac{4}{11}.$$

Thus the desired best fitting polynomial is

$$y = ax^2 + bx \qquad y = \frac{3}{11}x^2 - \frac{4}{11}x.$$

Recall from one variable calculus:

Suppose $\mathbf{z} = \mathbf{f}(\mathbf{x})$ and $\mathbf{x} = \mathbf{g}(t)$. Then we have the following **Chain Rule**:

$$\frac{d\mathbf{z}}{dt} = \frac{d\mathbf{z}}{d\mathbf{x}} \cdot \frac{d\mathbf{x}}{dt}.$$

with $\frac{d\mathbf{z}}{d\mathbf{x}}$ evaluated at $\mathbf{x} = \mathbf{g}(t)$.

Similarly for multi-variable calculus, we have

z dependent variable

x, y intermediate variables

t independent variable

Chain Rule - Case 1:

Suppose that $z = f(x, y)$ is a differentiable function of x and y , where $x = g(t)$ and $y = h(t)$ are both differentiable functions of t . Then

$$\frac{dz}{dt} = \frac{\partial f}{\partial x} \cdot \frac{dx}{dt} + \frac{\partial f}{\partial y} \cdot \frac{dy}{dt}$$

with $\frac{\partial f}{\partial x}$ and $\frac{\partial f}{\partial y}$ evaluated at $(x(t), y(t))$. Often this is abbreviated as:

$g(t), h(t)$

$$\boxed{\frac{dz}{dt} = \frac{\partial z}{\partial x} \cdot \frac{dx}{dt} + \frac{\partial z}{\partial y} \cdot \frac{dy}{dt}}$$

Justification: Recall the linear approximation (and upon writing $\Delta \mathbf{z} = \Delta f$):

$$\Delta \mathbf{z} = \Delta f \approx f_x(\mathbf{x}_o, y_o)\Delta x + f_y(\mathbf{x}_o, y_o)\Delta y.$$

Dividing both sides by Δt , we get

$$\frac{\Delta \mathbf{z}}{\Delta t} \approx f_x(\mathbf{x}_o, y_o)\frac{\Delta x}{\Delta t} + f_y(\mathbf{x}_o, y_o)\frac{\Delta y}{\Delta t}.$$

Letting $\Delta t \rightarrow 0$, we get

$$\frac{dz}{dt} = f_x(\mathbf{x}_o, y_o)\frac{dx}{dt} + f_y(\mathbf{x}_o, y_o)\frac{dy}{dt}.$$

As $(\mathbf{x}_o, y_o)$ is arbitrary, we get

$$\frac{dz}{dt} = \frac{\partial z}{\partial x} \frac{dx}{dt} + \frac{\partial z}{\partial y} \frac{dy}{dt}.$$

w : dependant variable
 x, y, z : intermediate variables
 t : indep. variable

For functions of three variables: $w = f(x, y, z)$, we similarly have corresponding linear approximation:

$$\Delta w \approx f_x(x_0, y_0, z_0)\Delta x + f_y(x_0, y_0, z_0)\Delta y + f_z(x_0, y_0, z_0)\Delta z$$

When x, y, z are functions of t , we also have corresponding Chain Rule:

$$\frac{dw}{dt} = \frac{\partial w}{\partial x} \cdot \frac{dx}{dt} + \frac{\partial w}{\partial y} \cdot \frac{dy}{dt} + \frac{\partial w}{\partial z} \cdot \frac{dz}{dt}.$$

Q09.1. Consider the function $w = 3x + xy + 2y^2$.

Suppose x is a function of t such that $\frac{dx}{dt} = e^{t^2}$ and

$y = e^t$. Find $\frac{dw}{dt}$, expressing your answer in terms of x

and t . Evaluate also $\left. \frac{dw}{dt} \right|_{t=2}$ if it is given that $x = 3$ when $t = 2$.

Solution: Given

$$w = 3x + xy + 2y^2, \quad \frac{dx}{dt} = e^{t^2}, \quad y = e^t.$$

By the Chain Rule,

$$\begin{aligned}\frac{dw}{dt} &= \frac{\partial w}{\partial x} \cdot \frac{dx}{dt} + \frac{\partial w}{\partial y} \cdot \frac{dy}{dt} \\ &= (3 + y) \cdot e^{t^2} + (x + 4y) \cdot e^t \\ &= (3 + e^t) \cdot e^{t^2} + (x + 4e^t) \cdot e^t \\ &= 3e^{t^2} + e^{t^2+t} + xe^t + 4e^{2t}.\end{aligned}$$

At $t = 2$, $x = 3$. So

$$\left. \frac{dw}{dt} \right|_{t=2} = 3e^4 + e^6 + 3e^2 + 4e^4 = e^6 + 7e^4 + 3e^2 \approx 807.78.$$

Q09.2. Consider the function $w = 2ye^x - \ln z$. Suppose

$$x = \ln(t^2 + 1), \quad y = \tan^{-1}(t), \quad z = e^t.$$

(a) Find $\frac{dw}{dt}$ by using the Chain Rule, and express your answer in terms of t .

(b) Find $\frac{dw}{dt}$ by first expressing w in terms of t .

w : dependent variable

x, y, z : intermediate variable

t : independent variable

Solution: (a) Given $w = 2ye^x - \ln z$, and

$$x = \ln(t^2 + 1), \quad y = \tan^{-1}(t), \quad z = e^t.$$

By the Chain Rule,

$$e^x = t^2 + 1$$

$$\begin{aligned} & \frac{dw}{dt} \\ &= \frac{\partial w}{\partial x} \cdot \frac{dx}{dt} + \frac{\partial w}{\partial y} \cdot \frac{dy}{dt} + \frac{\partial w}{\partial z} \cdot \frac{dz}{dt} \\ &= \underbrace{2ye^x} \cdot \frac{2t}{t^2 + 1} + \underbrace{2e^x} \cdot \frac{1}{1 + t^2} - \frac{1}{z} \cdot e^t \\ &= 2 \tan^{-1}(t) \cdot \cancel{(t^2 + 1)} \cdot \frac{2t}{\cancel{t^2 + 1}} + 2 \cancel{(t^2 + 1)} \cdot \frac{1}{\cancel{1 + t^2}} - \frac{1}{\cancel{e^t}} \cdot \cancel{e^t} \\ &= 4t \tan^{-1}(t) + 2 - 1 \\ &= 1 + 4t \tan^{-1}(t). \end{aligned}$$

(b) Given $w = 2ye^x - \ln z$, and

$$x = \ln(t^2 + 1), \quad y = \tan^{-1}(t), \quad z = e^t.$$

Thus

$$\begin{aligned} w &= 2ye^x - \ln z \\ &= 2 \cdot \tan^{-1}(t) \cdot (t^2 + 1) - t \\ &= \underline{2(t^2 + 1) \tan^{-1}(t) - t}. \end{aligned}$$

By product rule,

$$\begin{aligned} \underline{\frac{dw}{dt}} &= \frac{d}{dt}(2(t^2 + 1) \tan^{-1}(t) - t) \\ &= 4t \cdot \tan^{-1}(t) + 2(t^2 + 1) \cdot \frac{1}{1 + t^2} - 1 \\ &= 4t \tan^{-1}(t) + 2 - 1 \\ &= \underline{1 + 4t \tan^{-1}(t)}. \end{aligned}$$

Q09.3. Suppose the temperature T at any point in the xyz -space is given by

$$T(x, y, z) = e^x - yz + zx,$$

T dep. var.
 x, y, z inter. var.
 t indep. var.

measured in $^{\circ}\text{C}$. Here the coordinates x , y , z are measured in cm. If the position (x, y, z) of an insect at time t , measured in seconds, is given by

$$x = t, \quad y = 2(1 + t^2), \quad z = 3 + t,$$

find the rate of change of temperature experienced by the insect at time $t = 0$. Give your answer in $^{\circ}\text{C}$ per second.

$$\left. \frac{dT}{dt} \right|_{t=0}$$

Solution: Let $w = w(t)$ be the temperature experienced by the insect at time t . Then

$$w = T(x, y, z) = e^x - yz + zx$$

with $x = t$, $y = 2(1 + t^2)$, $z = 3 + t$.

Then by the Chain Rule,

$$\begin{aligned}\frac{dw}{dt} &= \frac{\partial T}{\partial x} \cdot \frac{dx}{dt} + \frac{\partial T}{\partial y} \cdot \frac{dy}{dt} + \frac{\partial T}{\partial z} \cdot \frac{dz}{dt} \\&= (e^x + z) \cdot 1 - z \cdot 4t + (-y + x) \cdot 1 \\&= e^t + (3 + t) - (3 + t) \cdot 4t - 2(1 + t^2) + t \\&= e^t - 6t^2 - 10t + 1.\end{aligned}$$

At $t = 0$, we have

$$\left. \frac{dw}{dt} \right|_{t=0} = e^0 - 0 - 0 + 1 = 2^\circ\text{C per second}.$$

Chain Rule - Case 2:

Suppose $z = z(x, y)$ and

$$x = x(u, v), \quad y = y(u, v).$$

Then regarding z as a function of u, v , we have

$$\frac{\partial z}{\partial u} = \frac{\partial z}{\partial x} \cdot \frac{\partial x}{\partial u} + \frac{\partial z}{\partial y} \cdot \frac{\partial y}{\partial u},$$

$$\frac{\partial z}{\partial v} = \frac{\partial z}{\partial x} \cdot \frac{\partial x}{\partial v} + \frac{\partial z}{\partial y} \cdot \frac{\partial y}{\partial v}.$$

Remark: The above formulas follow from Chain Rule for Case 1, since taking $\frac{\partial}{\partial u}$ is simply taking $\frac{d}{du}$ upon keeping v temporarily constant, etc.

z : dependent variable

x, y : intermediate variables

u, v : independent variables

$$z = z(u, v)$$

$$\frac{\partial z}{\partial u}, \frac{\partial z}{\partial v}$$

Similarly, suppose $w = w(x, y, z)$ and

$$x = x(u, v), \quad y = y(u, v), \quad , \quad z = z(u, v)$$

Then regarding w as a function of u, v , we have

$$\frac{\partial w}{\partial u} = \frac{\partial w}{\partial x} \cdot \frac{\partial x}{\partial u} + \frac{\partial w}{\partial y} \cdot \frac{\partial y}{\partial u} + \frac{\partial w}{\partial z} \cdot \frac{\partial z}{\partial u},$$

$$\frac{\partial w}{\partial v} = \frac{\partial w}{\partial x} \cdot \frac{\partial x}{\partial v} + \frac{\partial w}{\partial y} \cdot \frac{\partial y}{\partial v} + \frac{\partial w}{\partial z} \cdot \frac{\partial z}{\partial v}.$$

Q09.4. Consider the function $z = 4e^x \ln y$. Suppose

$$x = \ln(u \cos v), \quad y = u \sin v.$$

Use the Chain Rule to find $\frac{\partial z}{\partial u}$ and $\frac{\partial z}{\partial v}$, and express your answers in terms of u and v . Evaluate also $\frac{\partial z}{\partial u}$ and $\frac{\partial z}{\partial v}$ at the point $(u, v) = (2, \frac{\pi}{4})$.

z

x, y

u, v

Solution: Given $z = 4e^x \ln y$ with

$$x = \ln(u \cos v) = \ln u + \ln(\cos v), \quad y = u \sin v.$$

By the Chain Rule,

$$\frac{\partial x}{\partial u} = \frac{1}{u}$$

$$e^x = u \cos v$$

$$\frac{\partial z}{\partial u} = \frac{\partial z}{\partial x} \cdot \frac{\partial x}{\partial u} + \frac{\partial z}{\partial y} \cdot \frac{\partial y}{\partial u}$$

$$= 4e^x \ln y \cdot \frac{1}{u} + \frac{4e^x}{y} \cdot \sin v$$

$$= 4 \cancel{u} \cos v \cdot \ln(u \sin v) \cdot \frac{1}{\cancel{u}} + \frac{4 \cancel{u} \cos v}{\cancel{u \sin v}} \cdot \cancel{\sin v}$$

$$= 4 \cos v \cdot [\ln(u \sin v) + 1].$$

$$\left. \frac{\partial z}{\partial u} \right|_{(2, \pi/4)} = 4 \cos \frac{\pi}{4} \cdot \left(\ln\left(2 \sin \frac{\pi}{4}\right) + 1 \right)$$

$$= \underline{2\sqrt{2} \cdot (\ln(\sqrt{2}) + 1)} \approx 3.81.$$

Given $z = 4e^x \ln y$ with

$$x = \ln(u \cos v) = \ln u + \ln(\cos v), \quad y = u \sin v.$$

By the Chain Rule,

$$\begin{aligned} \frac{\partial z}{\partial v} &= \frac{\partial z}{\partial x} \cdot \frac{\partial x}{\partial v} + \frac{\partial z}{\partial y} \cdot \frac{\partial y}{\partial v} \\ &= 4e^x \ln y \cdot \left(\frac{-\sin v}{\cos v} \right) + \frac{4e^x}{y} \cdot (u \cos v) \\ &= 4u \cos v \cdot \ln(u \sin v) \cdot \left(\frac{-\sin v}{\cos v} \right) + \frac{4u \cos v}{u \sin v} \cdot (u \cos v) \\ &= -4u \sin v \cdot \ln(u \sin v) + \frac{4u \cos^2 v}{\sin v}. \end{aligned}$$

$$\begin{aligned} \left. \frac{\partial z}{\partial v} \right|_{(2, \pi/4)} &= -8 \cdot \frac{\sqrt{2}}{2} \cdot \ln(\sqrt{2}) + 8 \left(\frac{\sqrt{2}}{2} \right)^2 \cdot \sqrt{2} \\ &= -4\sqrt{2} \ln \sqrt{2} + 4\sqrt{2} \approx 3.70. \end{aligned}$$

Q09.5. A distant planet, Planet ***X***, orbits about a sun, and is orbited by a moon. The amount of rainfall on Planet ***X*** is given by

$$r = f(\theta, h) = h e^{-\frac{1}{\theta^2}},$$

r
 θ, h
 s, m

where θ is the average surface temperature on Planet ***X*** and h is the average humidity on the surface of Planet ***X***.

The quantities θ and h are also known to depend on the distance ***s*** of Planet ***X*** from its sun, and the distance ***m*** of Planet ***X*** from its moon. Suppose that when $\theta = 80$ and $h = 10$, you measured that

$$\frac{\partial \theta}{\partial s} = 10, \quad \frac{\partial \theta}{\partial m} = 0.1, \quad \frac{\partial h}{\partial s} = 5, \quad \text{and} \quad \frac{\partial h}{\partial m} = 0.5.$$

Calculate $\frac{\partial r}{\partial s}$ when $\theta = 80$ and $h = 10$.

Solution: Given $r = f(\theta, h) = he^{-\frac{1}{\theta^2}}$ with

r
 θ, h
 s, m

$$\theta = \theta(s, m), \quad h = h(s, m).$$

By the Chain Rule,

$$\begin{aligned} \frac{\partial r}{\partial s} &= \frac{\partial r}{\partial \theta} \cdot \frac{\partial \theta}{\partial s} + \frac{\partial r}{\partial h} \cdot \frac{\partial h}{\partial s} \\ &= \underline{he^{-\frac{1}{\theta^2}}} \cdot \frac{2}{\theta^3} \cdot \frac{\partial \theta}{\partial s} + e^{-\frac{1}{\theta^2}} \cdot \frac{\partial h}{\partial s}. \end{aligned}$$

Given that when $\theta = 80$ and $h = 10$,

$$\frac{\partial \theta}{\partial s} = 10, \quad \frac{\partial \theta}{\partial m} = 0.1, \quad \frac{\partial h}{\partial s} = 5, \quad \text{and} \quad \frac{\partial h}{\partial m} = 0.5.$$

$$\begin{aligned} \therefore \left. \frac{\partial r}{\partial s} \right|_{(\theta, h) = (80, 10)} &= \frac{10 \cdot 2}{80^3} e^{-\frac{1}{80^2}} \cdot 10 + e^{-\frac{1}{80^2}} \cdot 5 \\ &= \left(\frac{1}{2560} + 5 \right) e^{-\frac{1}{6400}} \approx 4.9996.. \end{aligned}$$

u
 x, y
 r, θ

Q09.6. Suppose $u = f(x, y)$ is a differentiable function on $\mathbb{R}^2$ and all its first-order and second-order partial derivatives are continuous, and $x = r \cos \theta$, $y = r \sin \theta$. Show that

$$\frac{\partial^2 u}{\partial r^2} = f_{xx} \cos^2 \theta + 2f_{xy} \cos \theta \sin \theta + f_{yy} \sin^2 \theta.$$

[Remark: First find $\frac{\partial u}{\partial r}$ in terms of f_x , f_y and θ . You may also use Clairaut's Theorem freely here.]

Soln: Given $u = f(x, y)$ with $x = r \cos \theta$, $y = r \sin \theta$.

By the Chain Rule, we have

$$\frac{\partial u}{\partial r} = \frac{\partial f}{\partial x} \cdot \frac{\partial x}{\partial r} + \frac{\partial f}{\partial y} \cdot \frac{\partial y}{\partial r} = f_x \cdot \cos \theta + f_y \cdot \sin \theta.$$

Handwritten notes: f_x u
 x, y x, y
 r, θ r, θ

$$\frac{\partial^2 u}{\partial r^2}$$

$$= \frac{\partial}{\partial r} \left(\frac{\partial u}{\partial r} \right) = \frac{\partial}{\partial r} (f_x \cdot \cos \theta + f_y \cdot \sin \theta)$$

$$= \frac{\partial}{\partial r} (f_x) \cdot \cos \theta + \frac{\partial}{\partial r} (f_y) \cdot \sin \theta$$

$$= (f_{xx} \cdot \frac{\partial x}{\partial r} + f_{xy} \cdot \frac{\partial y}{\partial r}) \cdot \cos \theta + (f_{yx} \cdot \frac{\partial x}{\partial r} + f_{yy} \cdot \frac{\partial y}{\partial r}) \cdot \sin \theta$$

$$= (f_{xx} \cos \theta + f_{xy} \sin \theta) \cos \theta + (f_{yx} \cos \theta + f_{yy} \sin \theta) \sin \theta$$

$$= f_{xx} \cos^2 \theta + 2f_{xy} \sin \theta \cos \theta + f_{yy} \sin^2 \theta.$$

In the last line, we used Clairaut's Theorem: $f_{xy} = f_{yx}$.

For a function $f(\mathbf{x}, \mathbf{y})$, the gradient of $f(\mathbf{x}, \mathbf{y})$ is the vector-valued function

$$\nabla f(\mathbf{x}, \mathbf{y}) = \langle f_x, f_y \rangle = f_x \mathbf{i} + f_y \mathbf{j} = \frac{\partial f}{\partial \mathbf{x}} \mathbf{i} + \frac{\partial f}{\partial \mathbf{y}} \mathbf{j} = \begin{pmatrix} f_x \\ f_y \end{pmatrix}$$

∇f is also read as 'del f '.

Similarly, for a function $f(\mathbf{x}, \mathbf{y}, \mathbf{z})$, its gradient is:

$$\nabla f(\mathbf{x}, \mathbf{y}, \mathbf{z}) = \langle f_x, f_y, f_z \rangle = f_x \mathbf{i} + f_y \mathbf{j} + f_z \mathbf{k} = \begin{pmatrix} f_x \\ f_y \\ f_z \end{pmatrix}$$

Q09.7. (a) Find the gradient (i.e. ∇f) of the function $f(x, y) = \sqrt{2x + 3y}$ at the point $(-1, 2)$.

(b) Consider the function

$$f(x, y, z) = \frac{1}{\sqrt{x^2 + y^2 + z^2}} + \ln(xyz).$$

Find ∇f at the point $(-1, 2, -2)$.

Solution: (a) Given $f(x, y) = \sqrt{2x + 3y}$.

$$\begin{aligned}\nabla f = f_x \mathbf{i} + f_y \mathbf{j} &= \frac{1}{2\sqrt{2x + 3y}} \cdot 2\mathbf{i} + \frac{1}{2\sqrt{2x + 3y}} \cdot 3\mathbf{j} \\ &= \frac{1}{\sqrt{2x + 3y}} \mathbf{i} + \frac{3}{2\sqrt{2x + 3y}} \mathbf{j}.\end{aligned}$$

At $(-1, 2)$,

$$\nabla f \Big|_{(-1, 2)} = \frac{1}{\sqrt{2(-1) + 3 \cdot 2}} \mathbf{i} + \frac{3}{2\sqrt{2(-1) + 3 \cdot 2}} \mathbf{j} = \frac{1}{2} \mathbf{i} + \frac{3}{4} \mathbf{j}.$$

Given

$$f(x, y, z) = (x^2 + y^2 + z^2)^{-1/2} + \ln x + \ln y + \ln z.$$

$$\nabla f = f_x \mathbf{i} + f_y \mathbf{j} + f_z \mathbf{k}$$

$$\begin{aligned} &= \left(-\frac{1}{2} \cdot (x^2 + y^2 + z^2)^{-3/2} \cdot 2x + \frac{1}{x} \right) \mathbf{i} \\ &\quad + \left(-\frac{1}{2} \cdot (x^2 + y^2 + z^2)^{-3/2} \cdot 2y + \frac{1}{y} \right) \mathbf{j} \\ &\quad + \left(-\frac{1}{2} \cdot (x^2 + y^2 + z^2)^{-3/2} \cdot 2z + \frac{1}{z} \right) \mathbf{k} \end{aligned}$$

$$\nabla f = -\frac{x\mathbf{i} + y\mathbf{j} + z\mathbf{k}}{(x^2 + y^2 + z^2)^{3/2}} + \frac{\mathbf{i}}{x} + \frac{\mathbf{j}}{y} + \frac{\mathbf{k}}{z}.$$

Then

$$\nabla f \Big|_{(-1,2,-2)} = -\frac{-\mathbf{i} + 2\mathbf{j} - 2\mathbf{k}}{27} - \mathbf{i} + \frac{\mathbf{j}}{2} - \frac{\mathbf{k}}{2} = -\frac{26}{27}\mathbf{i} + \frac{23}{54}\mathbf{j} - \frac{23}{54}\mathbf{k}.$$